

Introduction to mathematics for economists

Exercise Sheet #5 — Mathcamp 2026, Sciences Po Master's in Economics

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This sheet contains one short warm-up followed by three long, case-heavy problems rather than many small ones. You need to complete all 3 exercises.

Warm-up

Tip for KKT constraints

Throughout, write every constraint in the form $g_j(x) \leq 0$,

$$\max_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad g_j(x) \leq 0, \quad j = 1, \dots, J,$$

and build the Lagrangian by subtracting each constraint with a non-negative multiplier:

$$\mathcal{L}(x, \mu) = f(x) - \sum_{j=1}^J \mu_j g_j(x), \quad \mu_j \geq 0.$$

The KKT conditions are

- **Stationarity:** $\nabla f(x^*) = \sum_j \mu_j \nabla g_j(x^*)$;
- **Primal feasibility:** $g_j(x^*) \leq 0$ for all j ;
- **Dual feasibility:** $\mu_j \geq 0$ for all j ;
- **Complementary slackness:** $\mu_j g_j(x^*) = 0$ for all j .

A household splits a budget $B > 0$ between two goods and maximises

$$U(x_1, x_2) = \ln x_1 + \ln x_2$$

subject to $x_1 + x_2 \leq B$ and a **subsistence floor** $x_1 \geq m$, with $0 < m < B$.

Tasks.

- Solve the problem. For which values of m does the floor bind?
- Compute the multiplier λ on the budget constraint in each regime and verify that it is continuous at the switching point.
- Write the value function $U^*(B, m)$ explicitly in each regime and verify the envelope theorem by direct differentiation.

Solution

(a) Setup and regimes. The feasible set $\{x_1 \geq m, x_2 \geq 0, x_1 + x_2 \leq B\}$ is non-empty (since $m < B$), compact and convex; U is strictly concave on the positive orthant. So KKT are necessary and sufficient, and the solution is unique.

Since U is strictly increasing, the budget binds: $x_1 + x_2 = B$ and $\lambda > 0$. Writing $g_0 = x_1 + x_2 - B \leq 0$ and $g_1 = m - x_1 \leq 0$ with multipliers $\lambda, \mu \geq 0$,

$$\begin{aligned}\mathcal{L} &= \ln x_1 + \ln x_2 - \lambda(x_1 + x_2 - B) - \mu(m - x_1), \\ \frac{1}{x_1} - \lambda + \mu &= 0, \quad \frac{1}{x_2} - \lambda = 0, \quad \mu(x_1 - m) = 0.\end{aligned}$$

Regime I (floor slack, $\mu = 0$). Then $1/x_1 = 1/x_2 = \lambda$, so $x_1 = x_2 = B/2$ and $\lambda = 2/B$. Valid iff $x_1 \geq m$, i.e. $m \leq B/2$.

Regime II (floor binds, $\mu > 0$). Then $x_1 = m$, $x_2 = B - m$, $\lambda = 1/(B - m)$ and

$$\mu = \lambda - \frac{1}{m} = \frac{1}{B - m} - \frac{1}{m} = \frac{2m - B}{m(B - m)} \geq 0 \iff m \geq B/2.$$

The two conditions are exact complements: no gap, no overlap.

(b) Continuity of the shadow price. $\lambda_I = 2/B$ and $\lambda_{II} = 1/(B - m)$. At the switching point $m = B/2$, $\lambda_{II} = 1/(B/2) = 2/B = \lambda_I$. The shadow price of the budget is continuous even though the regime changes. The *allocation* is continuous too, but kinked: flat at $x_1 = B/2$, then rising one-for-one with m .

(c) Envelope theorem.

$$U^* = \begin{cases} 2 \ln(B/2), & m \leq B/2, \\ \ln m + \ln(B - m), & m \geq B/2. \end{cases}$$

Direct differentiation gives $\partial U^*/\partial B = 2/B$ in Regime I and $1/(B - m)$ in Regime II: exactly λ in each case. Similarly $\partial U^*/\partial m = 0$ in Regime I and $1/m - 1/(B - m) = -\mu$ in Regime II: **the marginal welfare cost of tightening the floor is exactly its multiplier.** A slack constraint costs nothing; a binding one costs μ per unit.

Public Investment under Political and Budget Constraints

A city chooses expenditures x_1 (transport) and x_2 (environment) to maximize

$$U(x_1, x_2) = \theta_1 \ln(1 + x_1) + \theta_2 \sqrt{x_2}, \quad \theta_1, \theta_2 > 0$$

where θ_1, θ_2 are weights estimated from a pre-election survey (think of them as capturing how much voters care about each policy area). The city faces a budget constraint $x_1 + x_2 \leq B$, and minimum funding mandates inherited from past legislation: $x_1 \geq m_1 > 0$, $x_2 \geq m_2 > 0$.

Tasks.

- Explain why the following assumption is needed $m_1 + m_2 < B$, and what goes wrong at $m_1 + m_2 = B$ and at $m_1 + m_2 > B$.
- Verify that the KKT conditions are necessary and sufficient here, and argue that the budget constraint always binds.
- Write the KKT conditions and find the interior solution in terms of $(\theta_1, \theta_2, B, m_1, m_2)$.
- Show that the two mandates can never bind simultaneously, and derive the exact thresholds separating the three regimes. Express them in terms of the single statistic θ_1/θ_2 , and discuss how shifts in voter priorities move the city between regimes.
- Interpret the budget multiplier, and compute the marginal welfare cost of the transport mandate.

Solution

(a) Feasibility. The mandates force $x_1 + x_2 \geq m_1 + m_2$ while the budget forces $x_1 + x_2 \leq B$. If $m_1 + m_2 > B$ the feasible set is **empty**: past legislation has promised more than the city can pay, and there is nothing to optimise. If $m_1 + m_2 = B$ the feasible set is the **single point** (m_1, m_2) : the answer is forced, preferences are irrelevant, and Slater's condition fails. Only $m_1 + m_2 < B$ gives a genuine choice problem.

In real world policy, it can be common that the real question is not the optimum but the *feasibility frontier*: how much discretion do accumulated commitments leave?

(b) U is strictly concave on $\{x_1 > -1, x_2 > 0\}$: its Hessian is diagonal with entries $-\theta_1/(1+x_1)^2 < 0$ and $-\theta_2/(4x_2^{3/2}) < 0$. All three constraints are affine, hence convex. Therefore:

- the feasible set is convex, compact and non-empty, so a maximum exists (Weierstrass);
- the constraints are affine, so KKT are **necessary** with no further qualification (Slater also holds, since $m_1 + m_2 < B$);
- U concave and constraints convex, so KKT are **sufficient**;
- U *strictly* concave, so the maximiser is **unique**.

Marginal utilities are strictly positive everywhere on the feasible set, so the city always spends its entire budget:

$$x_1 + x_2 = B, \quad \lambda > 0.$$

Hence, we can restrict our attention to cases where budget constraint is binding.

(c) Write the constraints as

$$g_0 := x_1 + x_2 - B \leq 0, \quad g_1 := m_1 - x_1 \leq 0, \quad g_2 := m_2 - x_2 \leq 0,$$

with multipliers $\lambda, \mu, \nu \geq 0$. The Lagrangian is

$$\mathcal{L} = \theta_1 \ln(1 + x_1) + \theta_2 x_2^{1/2} - \lambda(x_1 + x_2 - B) - \mu(m_1 - x_1) - \nu(m_2 - x_2).$$

Stationarity (note the $+\mu$ and $+\nu$):

$$\frac{\theta_1}{1+x_1} - \lambda + \mu = 0, \quad \frac{\theta_2}{2\sqrt{x_2}} - \lambda + \nu = 0.$$

Complementary slackness:

$$\lambda(B - x_1 - x_2) = 0, \quad \mu(x_1 - m_1) = 0, \quad \nu(x_2 - m_2) = 0,$$

with feasibility $x_1 \geq m_1$, $x_2 \geq m_2$, $x_1 + x_2 \leq B$.

Possible regimes. With $B > m_1 + m_2$, three regimes can arise:

- **Regime I:** interior with respect to mandates, $x_1 > m_1$, $x_2 > m_2$ ($\mu = \nu = 0$).
- **Regime II:** transport mandate binds, $x_1 = m_1$, $x_2 = B - m_1$ ($\mu > 0$, $\nu = 0$).
- **Regime III:** environment mandate binds, $x_2 = m_2$, $x_1 = B - m_2$ ($\nu > 0$, $\mu = 0$).

Regime I (interior). With $\mu = \nu = 0$, stationarity gives

$$\lambda = \frac{\theta_1}{1+x_1} = \frac{\theta_2}{2\sqrt{x_2}}, \quad x_1 + x_2 = B.$$

The first equality is the **equalisation of marginal utility per euro**: at an optimum the last euro must buy the same welfare wherever it is spent. Dividing the two FOCs,

$$1 = \frac{\theta_1}{1+x_1} \cdot \frac{2\sqrt{x_2}}{\theta_2} \quad \Leftrightarrow \quad 1+x_1 = 2\sqrt{x_2} \frac{\theta_1}{\theta_2} \quad \Leftrightarrow \quad x_1 = 2\sqrt{x_2} \frac{\theta_1}{\theta_2} - 1.$$

Substituting into the budget constraint, with $y \equiv \sqrt{x_2}$ and $a \equiv 2\theta_1/\theta_2$, gives the quadratic

$$y^2 + ay - (1+B) = 0,$$

whose only economically meaningful (positive) root is

$$y = \frac{-a + \sqrt{a^2 + 4(1+B)}}{2}.$$

Hence

$$x_2^* = y^2, \quad x_1^* = ay - 1, \quad \lambda^* = \frac{\theta_2}{2y} > 0.$$

(d) Suppose the 2 mandates bind at the same time. Then $x_1 = m_1$ and $x_2 = m_2$, so total spending would be

$$x_1 + x_2 = m_1 + m_2 < B,$$

by assumption in the setting. But part (b) established that the budget must bind, $x_1 + x_2 = B$. Contradiction. So at most one mandate binds, and the three regimes listed in (c) are the only possibilities.

Now, because the budget binds we may write $x_2 = B - x_1$ and treat x_1 as the only free variable $x_1 \in [m_1, B - m_2]$. Define the difference of marginal utilities of the two goods given x_1 :

$$\varphi(x_1) := \underbrace{\frac{\theta_1}{1+x_1}}_{\text{marginal utility of a euro on transport}} - \underbrace{\frac{\theta_2}{2\sqrt{B-x_1}}}_{\text{marginal utility of a euro on environment}}.$$

The first term is strictly decreasing in x_1 (as x_1 increases, "transport gets saturated"); the second is strictly increasing (the lesser the funding for environment, the more the

payoff for trading away transport funding towards environmental policy), so minus that term is strictly decreasing. Hence φ is **strictly decreasing** and has at most one zero. Economically, $\varphi(x_1)$ is the net gain from moving one euro from environment to transport; it falls as transport gets more funding, which is just diminishing returns.

Intuitively, we can see that $\varphi(x_1)$ determines whether you should increase funding to one or the other of the two policies: if $\varphi(x_1) > 0$, the marginal utility of transportation is greater than that of environmental policy so that an additional euro is better spent on transportation than on environment; conversely for $\varphi(x_1) < 0$. At $\varphi(x_1) = 0$, the municipality is at a fixed point: it has neither incentive to decrease x_1 to get more of x_2 nor incentives to decrease x_2 to get more of x_1 . This determines whether constraint bind!

- $\varphi(m_1) \leq 0$: even at the *lowest permitted* transport spending, the marginal euro still does more good in environment. The city would like to spend less than m_1 on transport, and only the floor stops it $\Rightarrow$ **Regime II**, $x_1 = m_1$.
- $\varphi(B - m_2) \geq 0$: even at the lowest permitted environment spending, the marginal euro still does more good in transport $\Rightarrow$ **Regime III**, $x_2 = m_2$.
- Otherwise $\varphi(m_1) > 0 > \varphi(B - m_2)$, so φ crosses zero strictly inside the interval $\Rightarrow$ **Regime I**. That crossing point is exactly the stationarity condition $\theta_1/(1 + x_1) = \theta_2/(2\sqrt{x_2})$ solved in (c).

We can now express these thresholds $\varphi(B - m_2)$ and $\varphi(m_1)$ in terms of θ_1/θ_2 :

$$\begin{aligned} \varphi(m_1) \geq 0 &\iff \frac{\theta_1}{1 + m_1} \geq \frac{\theta_2}{2\sqrt{B - m_1}} &\iff \frac{\theta_1}{\theta_2} \geq \rho_1 := \frac{1 + m_1}{2\sqrt{B - m_1}}, \\ \varphi(B - m_2) \leq 0 &\iff \frac{\theta_1}{1 + B - m_2} \leq \frac{\theta_2}{2\sqrt{m_2}} &\iff \frac{\theta_1}{\theta_2} \leq \rho_2 := \frac{1 + B - m_2}{2\sqrt{m_2}}. \end{aligned}$$

Hence, regime endpoint only depend on (i) the relative weights of transport and environment for voters and (ii) minimum funding inherited from the past mandate.

$$\frac{\theta_1}{\theta_2} \leq \rho_1 \Rightarrow \text{Regime II (transport mandate binds);}$$

$$\rho_1 < \frac{\theta_1}{\theta_2} < \rho_2 \Rightarrow \text{Regime I (interior);}$$

$$\frac{\theta_1}{\theta_2} \geq \rho_2 \Rightarrow \text{Regime III (environment mandate binds).}$$

Multipliers by regime:

$$\lambda_I = \frac{\theta_2}{2y}, \quad \lambda_{II} = \frac{\theta_2}{2\sqrt{B - m_1}}, \quad \lambda_{III} = \frac{\theta_1}{1 + B - m_2},$$

with $\mu_{II} = \lambda_{II} - \theta_1/(1 + m_1) \geq 0$ and $\nu_{III} = \lambda_{III} - \theta_2/(2\sqrt{m_2}) \geq 0$, each non-negative exactly on its own region.

Intepretation. As θ_1/θ_2 rises (voters care increasingly about transport relative to environment), the city moves through

transport mandate binds $\rightarrow$ interior $\rightarrow$ environment mandate binds.

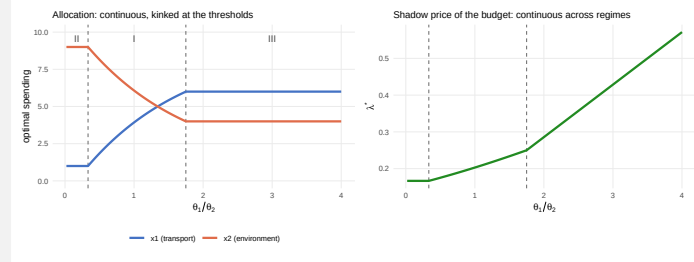
A mandate only “bites” for the policy area voters currently value *least*: when voters strongly favour environment (θ_1/θ_2 small), the transport floor m_1 becomes binding, because the city would otherwise want to spend even less than m_1 on transport. The

reverse holds symmetrically. Mandates can, in this precise sense, act as protections for minority priorities.

(e) Shadow prices. By the envelope theorem,

$$\frac{\partial U^*}{\partial B} = \lambda^* > 0, \quad \frac{\partial U^*}{\partial m_1} = -\mu^* \leq 0, \quad \frac{\partial U^*}{\partial m_2} = -\nu^* \leq 0.$$

So λ^* measures how much optimised voter welfare would rise if the city were allowed one more unit of budget. And μ^* is the welfare *cost* of the transport mandate: zero when slack, strictly positive when binding. Debates about repealing the mandates are mostly about one quantity, μ^* .



Agricultural Policy and Input Scarcity

A farm operator allocates land between wheat (w) and corn (c) to maximize estimated profit:

$$\hat{\pi}(w, c) = p_w w + p_c c, \quad p_w, p_c > 0$$

subject to (i) a **land constraint** $w + c \leq L$, (ii) a **fertilizer constraint** $a_w w + a_c c \leq F$ with $a_w, a_c > 0$, (iii) a **policy floor** $w \geq \alpha L$ (a minimum wheat-acreage mandate) with $\alpha \in [0, 1]$, and (iv) non-negativity $w, c \geq 0$. Parameters $(p_w, p_c, a_w, a_c, \alpha)$ are estimated from market data and agronomic studies. Across the exercise, we assume $F \geq a_w \alpha L$: there is enough fertilizer to meet the mandate. Assume $a_w < a_c$ and $p_w \neq p_c$.

Tasks.

- Reduce the problem by absorbing wheat policy floor into the objective. What are the *effective* land L' and fertilizer F' endowments?
- Before touching KKT, list the corners of the feasible set and explain why the optimum must be at one of them.
- Write the KKT conditions and classify the solution across all regimes. Which conditions on (p_w, p_c, a_w, a_c) select each corner?
- Explain the role of the effective scarcity ratio $F' L'$ and the role of α .

Solution

(a). Since the policy floor only sets a minimum for wheat, it is convenient to embed it directly into the objective and ask instead how much wheat is grown above the floor. Write

$$w = \alpha L + x, \quad x \geq 0,$$

where x is the extra wheat beyond the mandate. Substituting:

- Land: $(\alpha L + x) + c \leq L \Rightarrow x + c \leq L' := (1 - \alpha)L$.
- Fertilizer: $a_w(\alpha L + x) + a_c c \leq F \Rightarrow a_w x + a_c c \leq F' := F - a_w \alpha L$.

The objective becomes

$$\hat{\pi} = \underbrace{p_w \alpha L}_{\text{locked in}} + \underbrace{(p_w x + p_c c)}_{\text{variable part}},$$

so the reduced problem is

$$\max_{x, c \geq 0} p_w x + p_c c \quad \text{s.t.} \quad x + c \leq L', \quad a_w x + a_c c \leq F'. \quad (\star)$$

The effective endowments are L' (**free** land, after the mandate is served) and F' (**free** fertilizer).

(b). In our problem, everything is linear (from the objective to the constraints). In practice, the constraint space defines a polyhedron (defining it just in case you are introduced it in micro:): collect the constraints of $(\star)$ into matrix form. Writing $v = (x, c)^\top$,

$$P := \{v \in \mathbb{R}^2 : Av \leq b\}, \quad A = \begin{pmatrix} 1 & 1 \\ a_w & a_c \\ -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} L' \\ F' \\ 0 \\ 0 \end{pmatrix},$$

where the last two rows are the non-negativity constraints $-x \leq 0$ and $-c \leq 0$. Each row of $Av \leq b$ defines a closed half-plane, and P is the intersection of four of them, which is the definition of a polyhedron. It is nonempty ($(0, 0) \in P$ since $F' \geq 0 \rightarrow$ the mandate is fundable), closed, and bounded ($P \subseteq [0, L']^2$). Compact set + continuous objective, so by Weierstrass' theorem a maximiser exists.

Characterizing the vertices A vertex activates two of the four constraints; at most $\binom{4}{2} = 6$ candidates, and with $a_w \neq a_c$ at most four are feasible. With $f' := F'/L'$:

- $O = (0, 0)$: both floors active: wheat at exactly αL , no corn.
- $A = (\min\{L', F'/a_w\}, 0)$ and $B = (0, \min\{L', F'/a_c\})$: one crop only.
- $M = \left(\frac{a_c L' - F'}{a_c - a_w}, \frac{F' - a_w L'}{a_c - a_w}\right)$: both resources exhausted. Feasible **iff** $a_w < f' < a_c$.

Hence the three panels of the figure: triangle $\{(0, 0), (L', 0), (0, L')\}$ if $f' \geq a_c$; quadrilateral $\{(0, 0), (L', 0), M, (0, F'/a_c)\}$ if $a_w < f' < a_c$; triangle $\{(0, 0), (F'/a_w, 0), (0, F'/a_c)\}$ if $f' \leq a_w$.

Corner solutions. More importantly, the objective is strictly increasing. Hence, it is always optimal to hit at least one boundary. By contradiction, suppose both resource constraints were slack at some feasible (x, c) . Then for small $\varepsilon > 0$ the point $(x + \varepsilon, c)$ is still feasible and raises profit by $p_w \varepsilon > 0$. Hence, the constraint must bind.

Actually, we will show the solution must even be at a corner given our linear objective function. At each edge, substitute c . This gives $c = L' - x$ and $c = (F' - a_w x)/a_c$. We rewrite profit for corners of these 2 constraints:

$$p_c L' + (p_w - p_c)x, \quad \frac{p_c F'}{a_c} + a_w \left(\frac{p_w}{a_w} - \frac{p_c}{a_c} \right) x.$$

Each equation for profit is affine in x on a closed interval, hence maximised at an endpoint (unless prices / fertilizer needs equalize). Note: given our assumptions, we can ensure the solution must be at a corner, but more generally, with these kinds of linear programmes, the main result is "one of the vertex must be a solution" and not necessarily "any solution must be a vertex".

(c). Let $\lambda \geq 0$ be the multiplier on land, $\mu \geq 0$ on fertilizer, $\sigma \geq 0$ on $x \geq 0$ (equivalently on the mandate $w \geq \alpha L$), and $\nu \geq 0$ on $c \geq 0$. Writing the non-negativity constraints correctly as $-x \leq 0$ and $-c \leq 0$,

$$\mathcal{L} = p_w x + p_c c - \lambda(x + c - L') - \mu(a_w x + a_c c - F') + \sigma x + \nu c.$$

Stationarity (both floors give a plus sign):

$$p_w - \lambda - \mu a_w + \sigma = 0, \quad p_c - \lambda - \mu a_c + \nu = 0,$$

equivalently

$$\sigma = \lambda + \mu a_w - p_w \geq 0, \quad \nu = \lambda + \mu a_c - p_c \geq 0.$$

Complementary slackness: $\lambda(x + c - L') = 0$, $\mu(a_w x + a_c c - F') = 0$, $\sigma x = 0$, $\nu c = 0$.

Exploring the regimes. Since $p_w, p_c > 0$, at least one resource binds. Let $f' := F'/L'$ be the **fertilizer per unit of free land**. Assume for concreteness $a_c > a_w$ (corn is the thirstier crop); the other case is symmetric under relabelling.

Regime L: land-limited, $f' \geq a_c$. Fertilizer is abundant enough to plant *either* crop on all the free land, so the fertilizer constraint slacks: $\mu = 0$, $\lambda > 0$. Stationarity gives $\sigma = \lambda - p_w$ and $\nu = \lambda - p_c$, one of which must vanish, so $\lambda = \max\{p_w, p_c\}$:

- $p_w > p_c$: wheat only, $x = L'$, $\lambda = p_w$, $\nu = p_w - p_c > 0$, $\sigma = 0$.
- $p_c > p_w$: corn only, $c = L'$, $\lambda = p_c$, $\sigma = p_c - p_w > 0$, $\nu = 0$. **The mandate binds:** the farmer plants exactly αL of wheat and nothing more, and σ is the per-acre cost of the policy.
- $p_w = p_c$: knife-edge, any split of L' would be optimal. Solution not unique, multipliers not unique.

When land is the only scarce input, **profit per acre** p_i is the sole criterion.

Regime F: fertilizer-limited, $f' \leq a_w$. Land is abundant; the fertilizer constraint binds and land goes unused: $\lambda = 0$, $\mu > 0$. Then $\sigma = \mu a_w - p_w$ and $\nu = \mu a_c - p_c$, so $\mu = \max\{p_w/a_w, p_c/a_c\}$:

- $p_w/a_w > p_c/a_c$: wheat only, $x = F'/a_w$, $\mu = p_w/a_w$, $\nu = a_c(p_w/a_w - p_c/a_c) > 0$.
- $p_c/a_c > p_w/a_w$: corn only, $c = F'/a_c$, $\mu = p_c/a_c$, $\sigma = a_w(p_c/a_c - p_w/a_w) > 0$ — the mandate binds again.

When fertilizer is the only scarce input, the criterion is **profit per unit of the scarce input**, p_i/a_i . Important: the two criteria can disagree, and that is the whole point of the next regime.

Regime M (middle): both may bind, $a_w < f' < a_c$. The wheat-only vertex is land-limited at $(L', 0)$, the corn-only vertex is fertilizer-limited at $(0, F'/a_c)$, and the interior vertex M exists. If both crops are grown ($x, c > 0$) then $\sigma = \nu = 0$ and

$$\lambda + \mu a_w = p_w, \quad \lambda + \mu a_c = p_c,$$

a 2×2 linear system solved by

$$\mu = \frac{p_c - p_w}{a_c - a_w}, \quad \lambda = \frac{p_w a_c - p_c a_w}{a_c - a_w}.$$

Under $a_c > a_w$, dual feasibility requires **both**

$$\mu \geq 0 \iff p_c \geq p_w \quad \text{and} \quad \lambda \geq 0 \iff \frac{p_w}{a_w} \geq \frac{p_c}{a_c}.$$

So the farmer diversifies exactly when **corn wins per acre while wheat wins per unit of fertilizer**. If instead one crop wins on both criteria it is grown alone:

- $p_c > p_w$ and $p_c/a_c > p_w/a_w$: corn dominates, $c = F'/a_c$, fertilizer binds, land slack, mandate binds ($\sigma > 0$).
- $p_w > p_c$ and $p_w/a_w > p_c/a_c$: wheat dominates, $x = L'$, land binds, fertilizer slack.
- The fourth sign combination ($p_w > p_c$ and $p_c/a_c > p_w/a_w$) is **impossible** when $a_c > a_w$: the second inequality gives $p_c/a_c > p_w/a_w > p_w/a_c$, hence $p_c > p_w$, a contradiction.

Knife-edges. If $p_w = p_c$ in Regime L, or $p_w/a_w = p_c/a_c$ in Regime F, the objective is constant along an entire edge and any split is optimal. If $f' = a_w$ exactly, the vertex $(L', 0)$ makes both constraints bind at once: the vertex is *degenerate* and the multipliers are no longer unique: a whole interval of (λ, μ) satisfies KKT.

Scarcity	Criterion	Winner grown
$f' \geq a_c$: land only	profit per acre p_i	higher p_i
$f' \leq a_w$: fertilizer only	profit per unit fert. p_i/a_i	higher p_i/a_i
$a_w < f' < a_c$: both may bind	both criteria	both crops iff criteria disagree

(d). The allocation is driven by the **effective scarcity ratio** $f' = F'/L'$, compared against the two input intensities a_w, a_c ; and, conditional on the regime, by the two ranking criteria p_i and p_i/a_i .

Two consequences. First, the thresholds are *not* $\max\{a_w, a_c\}$ and $\min\{a_w, a_c\}$ in terms of f :

$$f' \geq a_w \iff f \geq a_w, \quad \text{but} \quad f' \geq a_c \iff f \geq (1 - \alpha)a_c + \alpha a_w,$$

so the wheat threshold is invariant to the mandate while the corn threshold is an α -weighted average of the two intensities. The mandate moves the boundary of the land-limited regime.

Second:

$$\frac{\partial f'}{\partial \alpha} = \frac{f - a_w}{(1 - \alpha)^2},$$

so tightening the wheat mandate makes the farm effectively more fertilizer-abundant whenever $f > a_w$. The intuition: acres forced into wheat consume fertilizer at rate a_w , below the farm-wide availability per acre, so they leave the *residual* land relatively better supplied. A policy aimed at wheat acreage therefore has a second-round effect on which constraint binds for everything else.

Anatomy of the feasible set: each inequality is a half-plane, P is what survives all four

